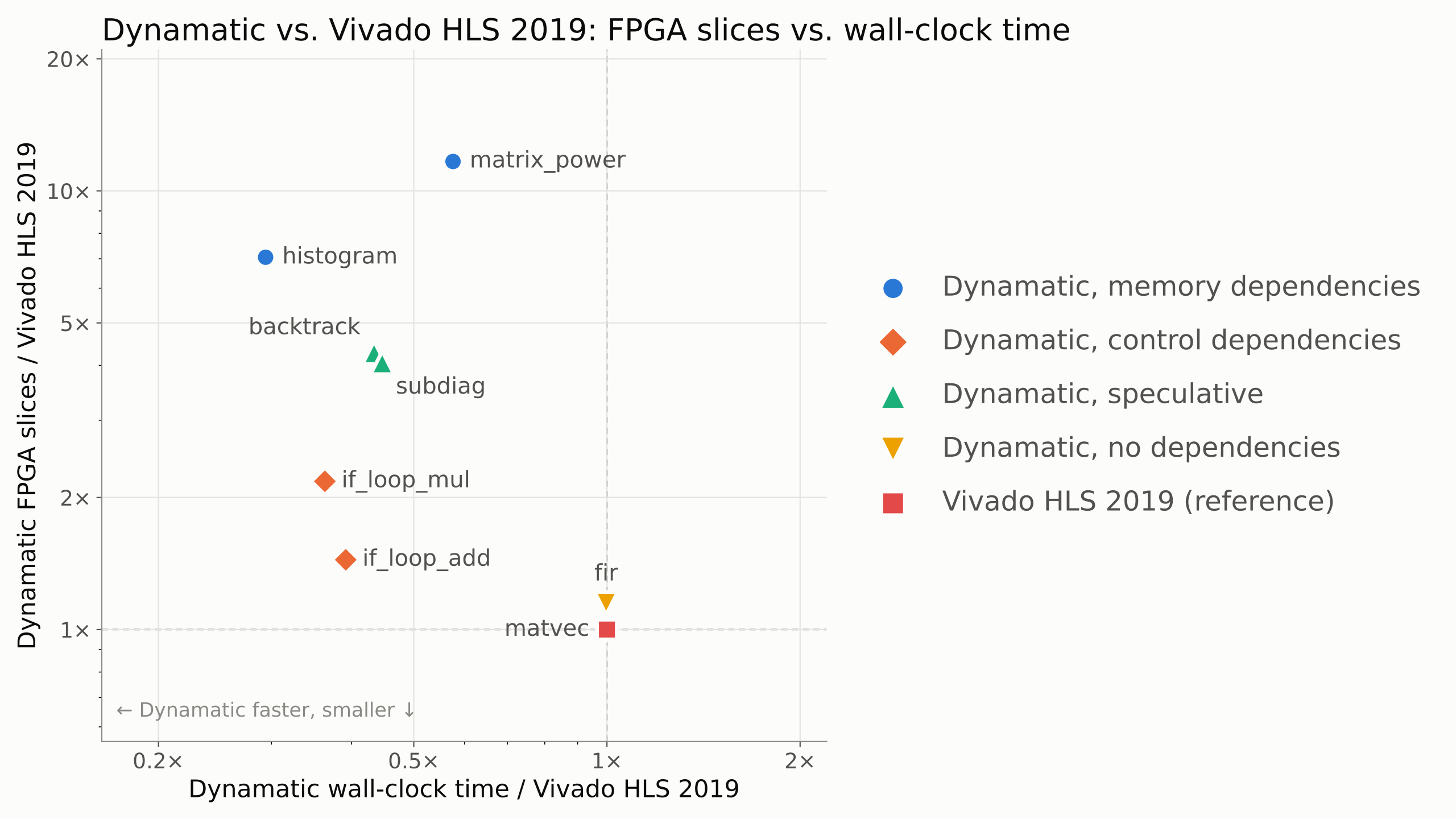
Dynamatic vs. Vivado HLS 2019: FPGA slices vs. wall-clock time
Dynamatic FPGA slices / Vivado HLS 2019
20×
10×
5×
2×
1×
matrix_power
histogram
backtrack
subdiag
if_loop_mul
if_loop_add
fir
matvec
← Dynamatic faster, smaller ↓
0.2×
0.5×
1×
2×
Dynamatic wall-clock time / Vivado HLS 2019
Dynamatic, memory dependencies
Dynamatic, control dependencies
Dynamatic, speculative
Dynamatic, no dependencies
Vivado HLS 2019 (reference)